

Quantum Generative Machine Learning

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Generative Machine Learning

Auto regressive

Diffusion

VAE

Generative Machine Learning

GAN

Flow based

Quantum

Quantum

Financial time series gen ^[1]

Drug design gen ^[2]

Image gen ^[3]

State gen ^[4,5]

Quantum

Financial time series gen ^[1]

Drug design gen ^[2]

Image gen ^[3]

State gen ^[4,5]

Problem Statement

- Consider a target distribution $\mathcal{E}_0$ of pure quantum states in a Hilbert space $\mathcal{H}$.
- The task is to generate new states from this distribution, given only a finite number of samples $\mathcal{S}_0 = \left\{ \left| \psi_i^{(0)} \right\rangle \right\}_{i=1}^m$.
- The model has to find the optimal parameters θ^* that minimize the distance between the generated and target distributions:

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\mathcal{E}_{\text{gen}(\theta)}, \mathcal{E}_0)$$

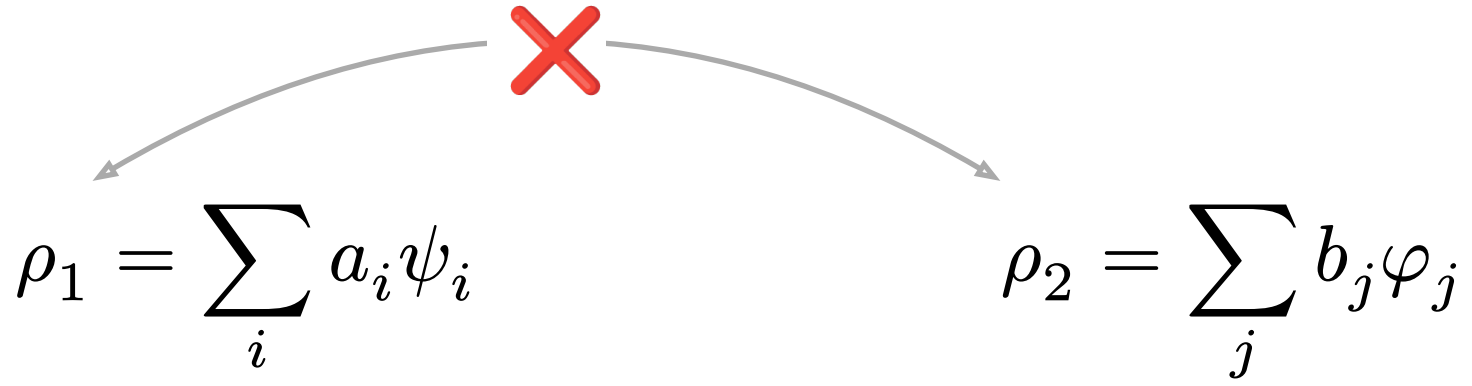
a **Distance** an
Metric **Optimizer**

a **Distance**

Metric

$$\rho_1 = \sum_i a_i \psi_i$$

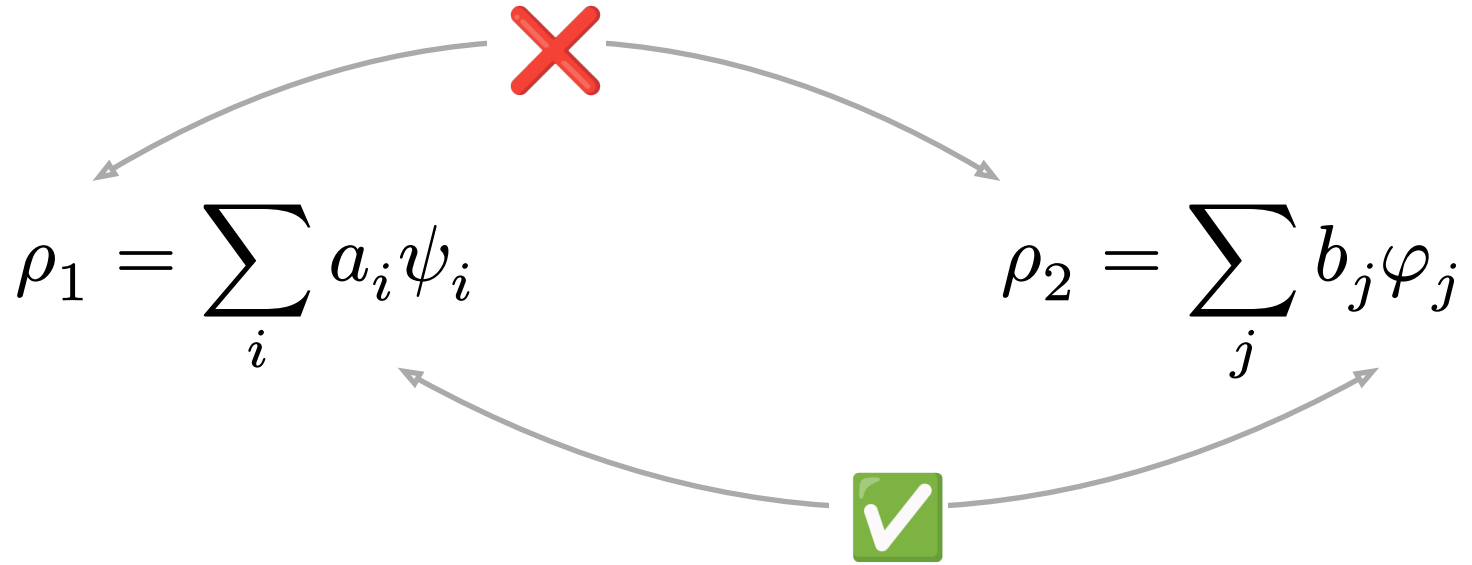
$$\rho_2 = \sum_j b_j \varphi_j$$



The diagram illustrates a distance metric between two probability density functions, ρ_1 and ρ_2 . The functions are defined as $\rho_1 = \sum_i a_i \psi_i$ and $\rho_2 = \sum_j b_j \varphi_j$. A large red 'X' is positioned above the space between the two equations. Two curved gray arrows originate from the 'X' and point towards each equation, indicating a relationship or distance between them.

$$\rho_1 = \sum_i a_i \psi_i$$
$$\rho_2 = \sum_j b_j \varphi_j$$

Distance Metrics



MAXIMUM MEAN DISCREPANCY (MMD)

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- Kernel based measure
- Compares the average features of two distributions
- $\mathcal{E}_1$ and $\mathcal{E}_2 \rightarrow$ two distributions of pure quantum states
 $\mathcal{S}_1$ and $\mathcal{S}_2 \rightarrow$ finite sets of sizes n_1 and n_2 sampled from these distributions
For $k := F(|\varphi\rangle, |\psi\rangle) = |\langle\varphi|\psi\rangle|^2$, the squared empirical MMD is defined as

$$\mathcal{D}_{\text{MMD}}^2(\mathcal{S}_1, \mathcal{S}_2) = \bar{F}(\mathcal{S}_1, \mathcal{S}_1) + \bar{F}(\mathcal{S}_2, \mathcal{S}_2) - 2\bar{F}(\mathcal{S}_1, \mathcal{S}_2)$$

where

$$\bar{F}(\mathcal{S}_1, \mathcal{S}_2) = \frac{1}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} F(|\varphi_i\rangle, |\psi_j\rangle),$$

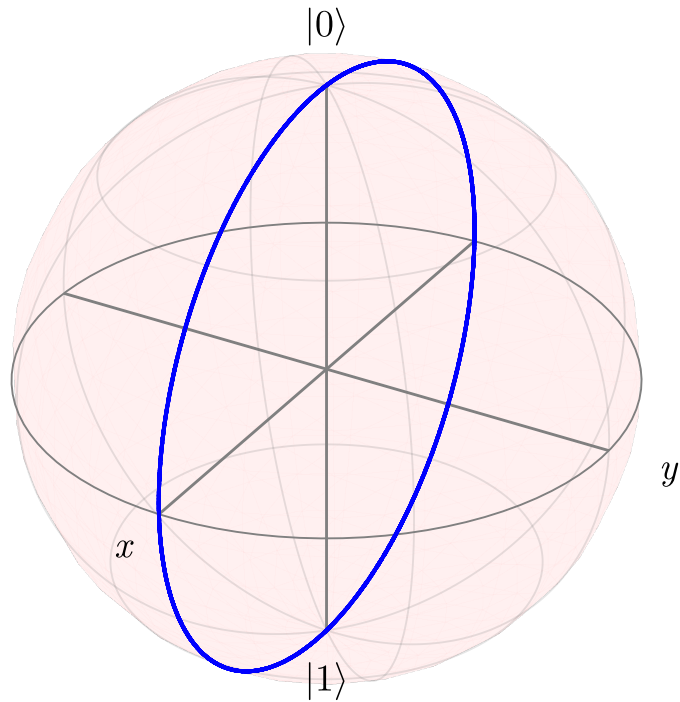


Figure 1: States lying on X-Z plane of the Bloch sphere.

$\approx$

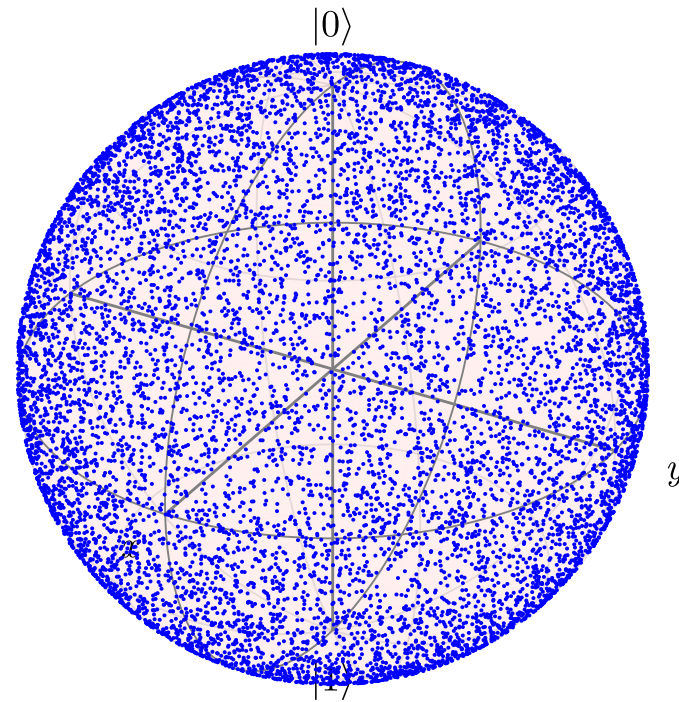


Figure 2: States sampled from a Haar random distribution.

1-WASSERSTEIN DISTANCE (W_1)

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- Compares two distributions. How much mass must be transported to transform one distribution into the other.
- A transport plan specifies how much probability mass from each sample of the first distribution is matched with each sample of the second.
- Minimize the total transport cost, so unlike MMD, Wasserstein distance uses the geometry of the sample space explicitly.

$$W_1(\mathcal{S}_1, \mathcal{S}_2) = \min_{\Gamma} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \Gamma_{ij} c(\varphi_i, \psi_j), \quad s.t. \quad \begin{aligned} \Gamma \mathbf{1}_{n_2} &= \frac{1}{n_1} \mathbf{1}_{n_1}, \\ \Gamma^T \mathbf{1}_{n_1} &= \frac{1}{n_2} \mathbf{1}_{n_2}, \\ \Gamma_{ij} &\geq 0. \end{aligned}$$

where $c(\varphi_i, \psi_j) = 1 - F(|\varphi_i\rangle, |\psi_j\rangle)$

an
Optimizer

Optimizers

- Need to find the optimal parameters such that the distance between the generated and target distributions is minimized.
- Optimizers can be broadly classified into two categories: gradient-based and gradient-free.

GRADIENT-BASED OPTIMIZERS

- Traverse the parameter space in the direction of steepest descent by computing the gradient of the chosen objective function with respect to the parameters.
- MMD $\rightarrow$ automatic differentiation.
- Wasserstein distance calculation is done through an **iterative algorithm**, and this makes it difficult for the autodiff engine to compute the gradient numerically because it has to track every single step of that loop to figure out the final gradient.

But there is an analytical simplification!

Envelope theorem

Let $f(x, \theta)$ and $l(x)$ be real-valued continuously differentiable functions, where $x \in \mathbb{R}^n$ are choice variables and $\theta \in \mathbb{R}^m$ are parameters. Denote x^* to be the optimal solution of f with constraint $l = 0$ and fixed θ , i.e.

$$x^* = \operatorname{argmin}_x f(x, \theta) \quad s.t. \quad l(x) = 0.$$

Then, assume that V is continuously differentiable function defined as $V(\theta) \equiv f(x^*(\theta), \theta)$, the derivative of V over parameters is

$$\frac{\partial V}{\partial \theta} = \frac{\partial f}{\partial \theta}$$

- $V(\theta)$ is the Wasserstein distance W_1 .

The choice variable x is the transport plan Γ .

The objective function is $f(\Gamma, \theta) = \langle \Gamma, C \rangle = \sum_{ij} C_{ij} \Gamma_{ij}$.

Substituting and solving:

$$\frac{\partial W}{\partial \theta} = - \sum_{ij} \Gamma_{ij}^* \frac{\partial}{\partial \theta} |\langle \varphi_i | \psi_j(\theta) \rangle|^2.$$

So the gradient of the Wasserstein distance can be calculated finding the optimal transport plan and the gradient of the pairwise Fidelity.

This gradient can be fed into any gradient-based optimizer.

GRADIENT-FREE OPTIMIZERS

GRADIENT-FREE OPTIMIZERS

- Rely on evaluating the objective function at different points in the parameter space to guide the search for optimal parameters.

Rotosolve

Rotosolve is a coordinate-wise optimization algorithm that iteratively optimizes one parameter at a time while keeping others fixed ^[6] using the following update rule for the d -th parameter:

$$\theta_d^* = -\frac{\pi}{2} - \arctan2\left(2\langle H \rangle_{\theta_d=0} - \langle H \rangle_{\theta_d=\pi/2} - \langle H \rangle_{\theta_d=-\pi/2}, \langle H \rangle_{\theta_d=\pi/2} - \langle H \rangle_{\theta_d=-\pi/2}\right)$$

where $\langle H \rangle_{\theta_d}$ denotes the expectation value of the objective function restricted to only depend on the parameter θ_d .

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But it's not useful for us!

- Only useful when the objective function can be represented as a Hermitian operator.
- We have a distance metric between distributions, and not an expectation value of a Hermitian operator.

Simultaneous Perturbation Stochastic Approximation (SPSA)

SPSA estimates the gradient by perturbing all parameters simultaneously in random directions^[7] with the following update rule:

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - a_k \frac{f(\boldsymbol{\theta}_k + c_k \boldsymbol{\Delta}_k) - f(\boldsymbol{\theta}_k - c_k \boldsymbol{\Delta}_k)}{2c_k} \begin{pmatrix} \Delta_{k1}^{-1} \\ \Delta_{k2}^{-1} \\ \vdots \\ \Delta_{kp}^{-1} \end{pmatrix}$$

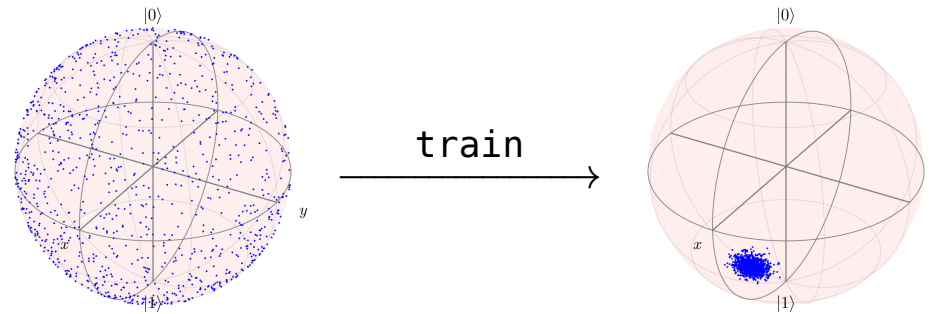
- Only two evaluations of the objective function per iteration are required, regardless of the number of parameters.
- However, it can be noisy and may require careful tuning of hyperparameters a_k and c_k .

Quantum Generative Model Architectures

for state generation

QUANTUM DIRECT TRANSPORT (QuDT)

- Learns a direct mapping from the initial distribution of quantum states to the target distribution.
- A single, deep Parameterized Quantum Circuit (PQC) transport the initial states to the target states. ^[5]



- **Toy problem:** Haar random ensemble $\rightarrow$ States clustered around any arbitrary state

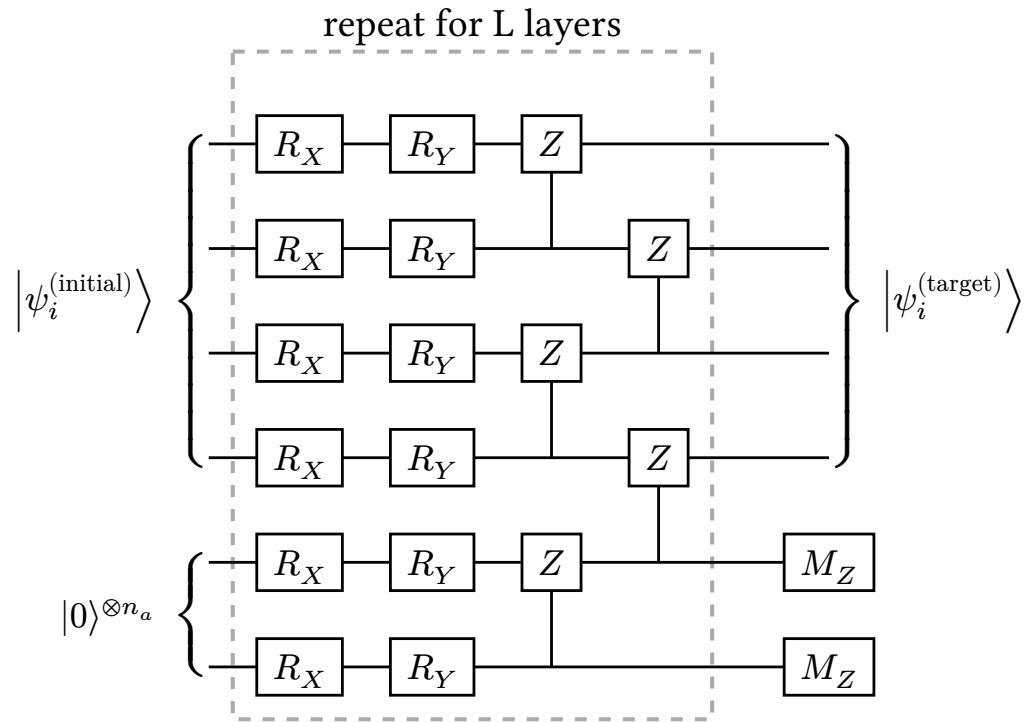


Figure 3: The QuDT circuit. M_Z represents measurement in the Z basis.

QUANTUM DDPM (QuDDPM)

- An ensemble of states is scrambled over multiple steps in the forward process and then a series of PQCs are trained to reverse this scrambling process step by step in the reverse process.
- This divides the task of learning a map from full noise to the target distribution into subtasks with low-depth circuits which can avoid barren plateaus.

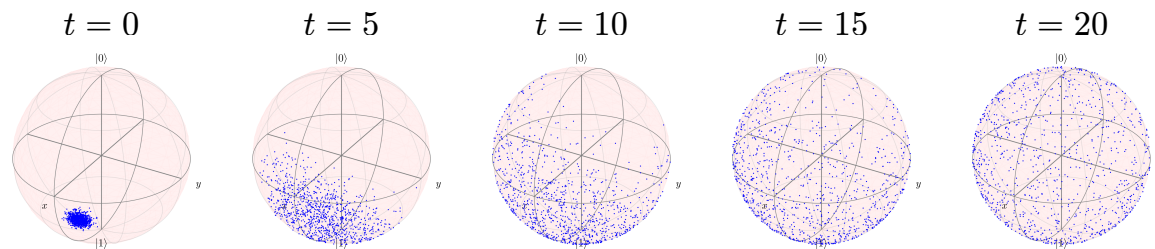
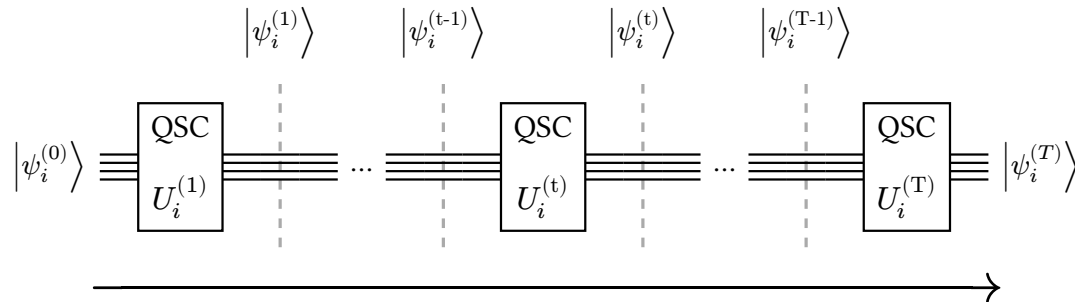
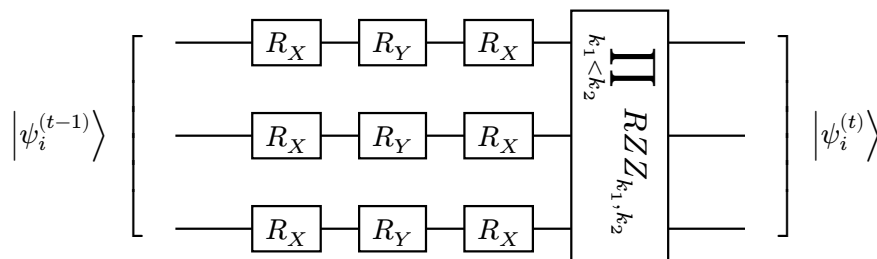


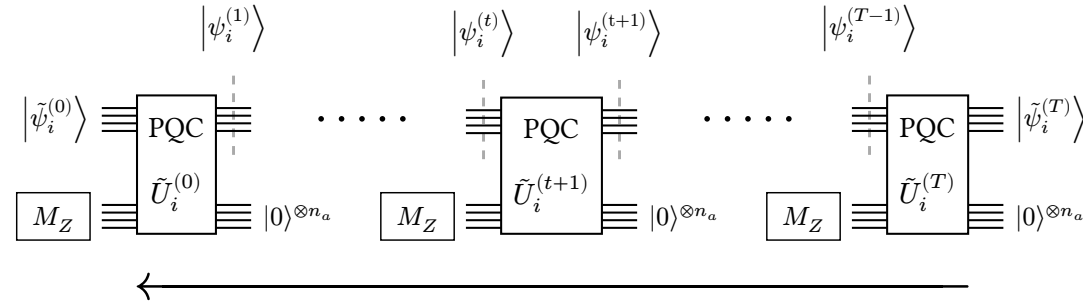
Figure 4: Scrambling $m = 1000$ states sampled from the toy distribution over $T = 20$ steps, as shown on the Bloch sphere.



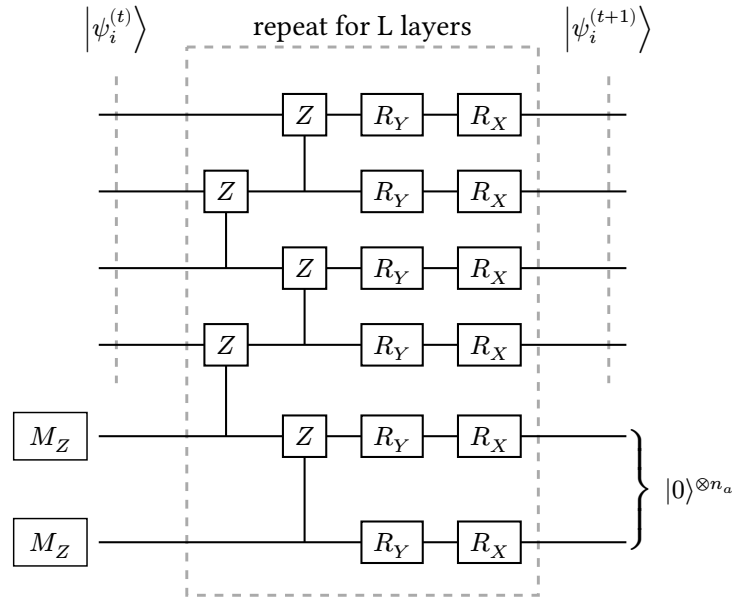
(a) Forward process circuit



(b) Quantum Scrambling Circuit (QSC) Block



(a) Reverse process circuit.



(b) PQC block (HEA) used in the reverse circuit.

- Measurements are necessary, as the denoising map is contractive and maintains the purity of all generated data in $\tilde{S}_0$.
- No specific constraint is placed on the measurement result. The measurement on ancillas is performed and discarded; only the post-measurement state of the data qubits is collected ^[5].



Connection with QuDT

The QuDT model can be seen as a special case of QuDDPM with $T = 1$ step in the reverse process, where the forward process is just a single random unitary that scrambles the initial distribution. This justifies the choice of ancillary qubits and their measurement in the QuDT model.

SEQUENTIAL QUANTUM DIRECT TRANSPORT (S-QuDT)

This hybrid model is proposed, taking ideas from both QuDT and QuDDPM.

- The model consists of T steps, where at each step t a PQC is trained to learn a direct transport map from the distribution at step t to the target distribution, instead of learning a denoising map to the less scrambled distribution as in QuDDPM.
- This removes the forward scrambling process entirely.

Results

CLUSTERED STATE GENERATION

- 4 qubit clustered states
- 2 ancilla qubits
- AMSGrad optimizer with learning rate 0.01
- $n = 1000$ samples
- Haar distribution $\rightarrow$ Clustered distribution
- Batch size = 200
- $T = 5$ steps

Results

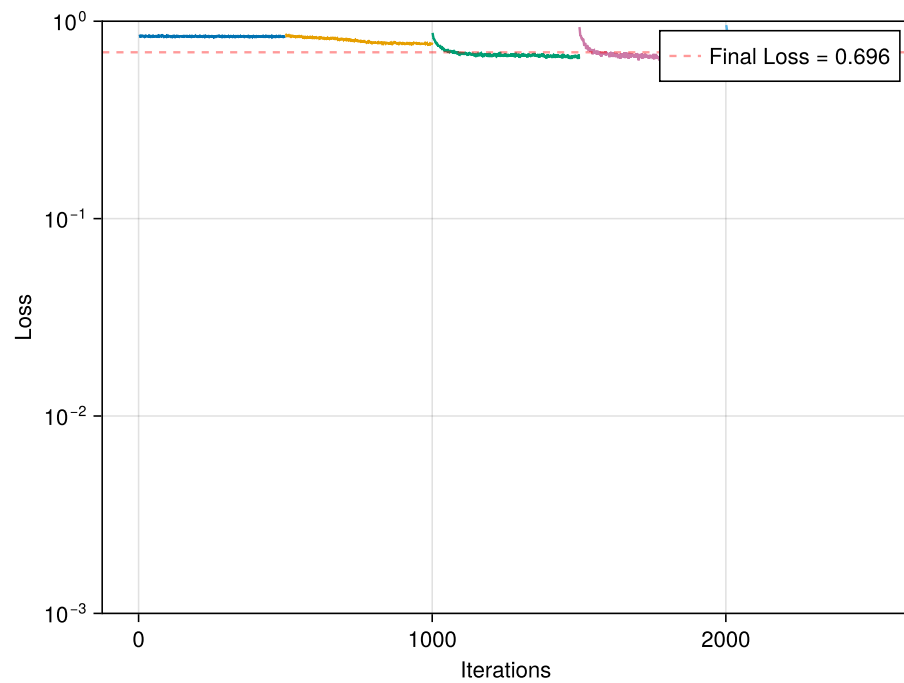


Figure 7: QuDDPM

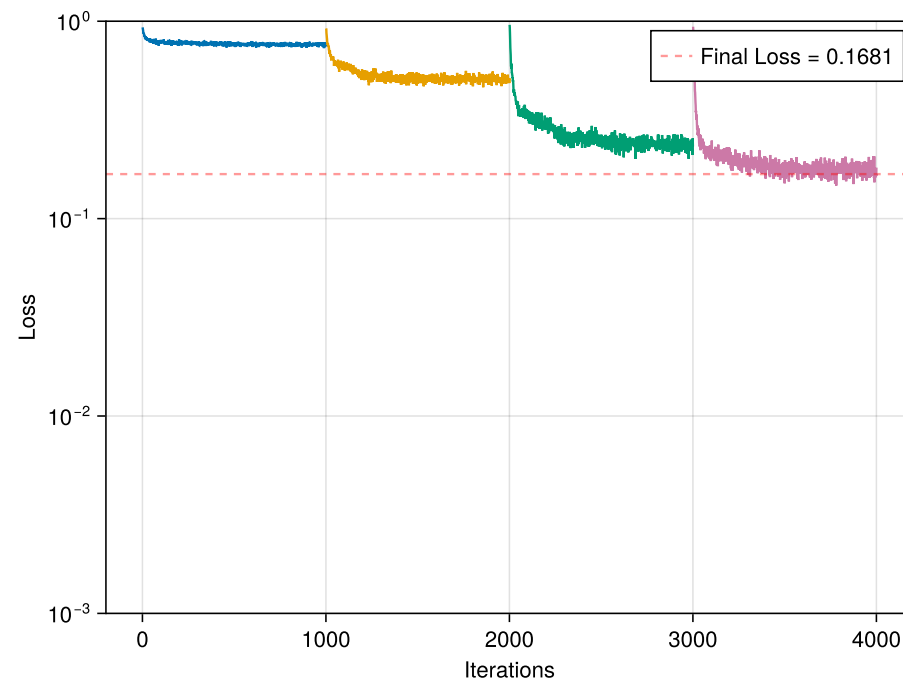


Figure 8: S-QuDT

HEA vs EHA

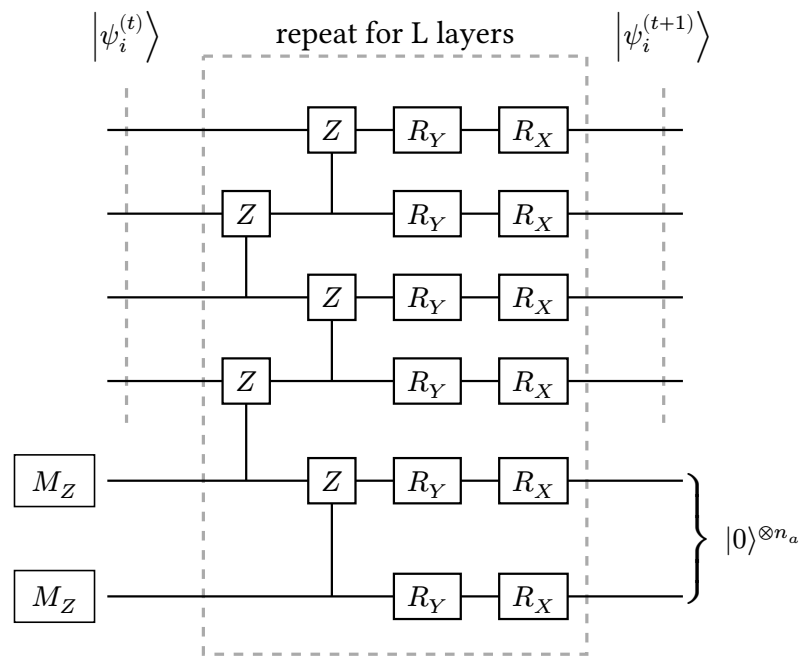


Figure 9: HEA block

VS

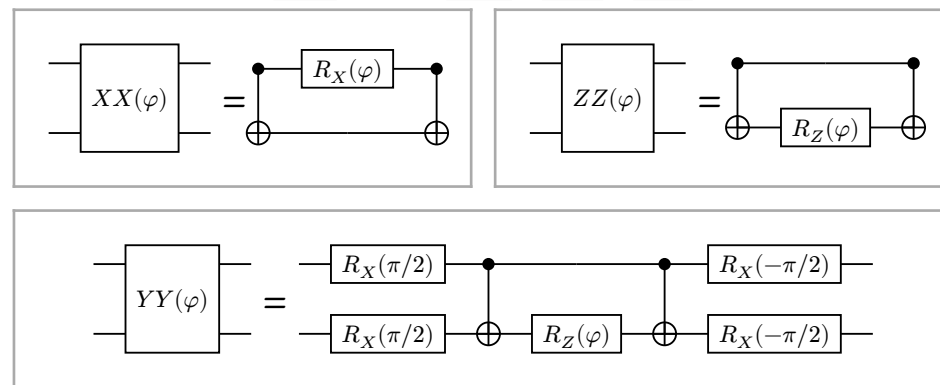
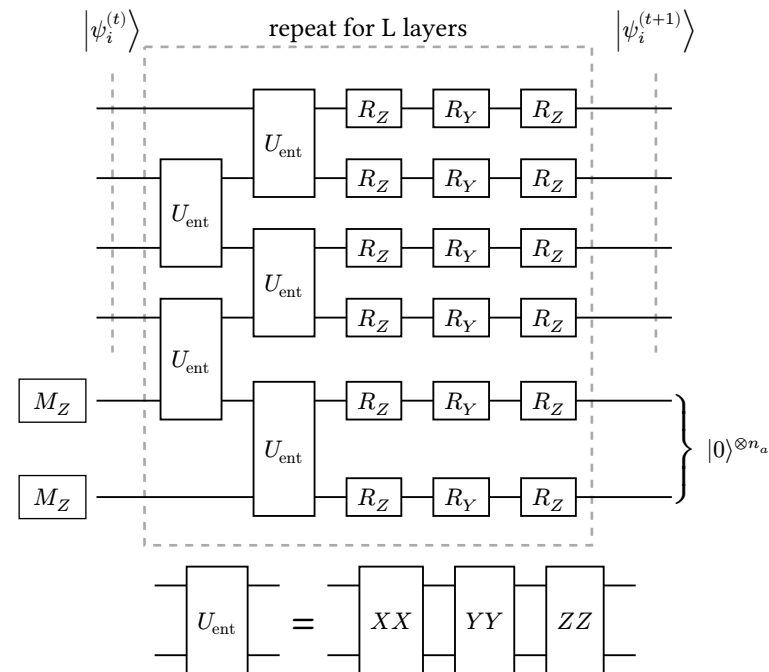


Figure 10: EHA block

TRANSVERSE FIELD ISING MODEL (TFIM) STATE GENERATION

- 9 qubit TFIM states
- 6 ancilla qubits
- AMSGrad optimizer with learning rate 0.02
- $n = 5000$ samples
- Haar distribution $\rightarrow$ TFIM ground states distribution
- Batch size = 400
- $T = 8$ steps
- 14 EHA layers per step

Results

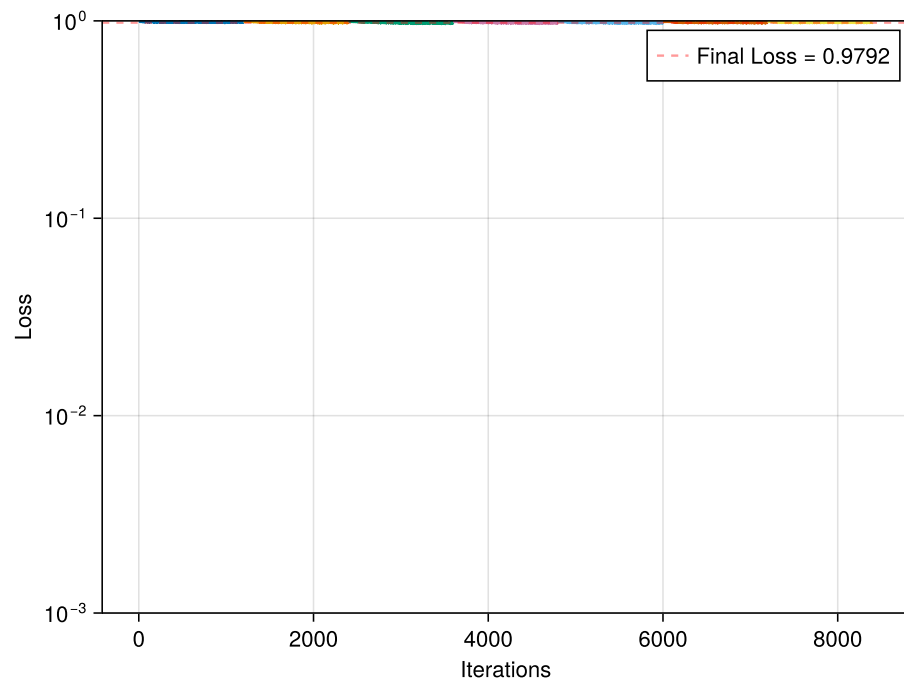


Figure 11: QuDDPM

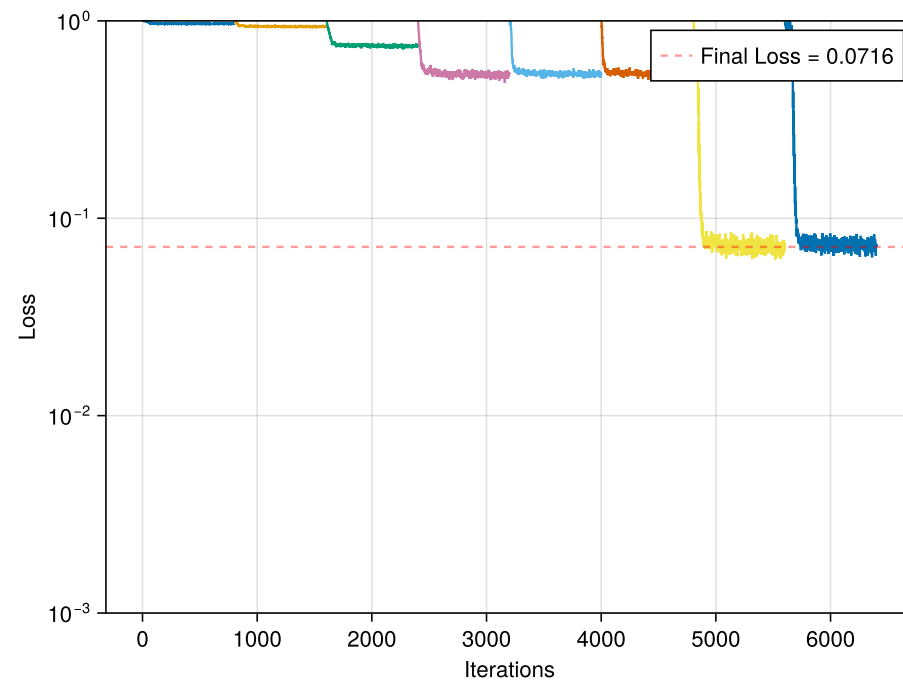


Figure 12: S-QuDT

Implementation

2k LoC of pure **Julia** with **Yao.jl**, with GPU acceleration using **CUDA.jl**.

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> , tokei -e archive/ -e '*.svg' -e '*.toml' -e '*.ipynb'
```

Language	Files	Lines	Code	Comments	Blanks
Julia	23	2189	1658	152	379
Markdown	5	25	0	19	6
Nix	10	281	243	22	16
Python	12	2130	1655	98	377
Shell	3	47	23	14	10
Typst	13	2664	2037	27	600
Total	66	7336	5616	332	1388

Conclusion & Future Work

- S-QuDT is more efficient and matches/outperforms QuDDPM, and completely eliminates the need for a forward scrambling process.
- EHA is a better ansatz than HEA, as it allows entanglement tuning.
- Both S-QuDT and QuDDPM fail to converge for exponentially localized states of Quantum Kicked Rotor, so there is scope of improvement.
- Potential to work on gradient-free optimizers for such a model.

Bibliography

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